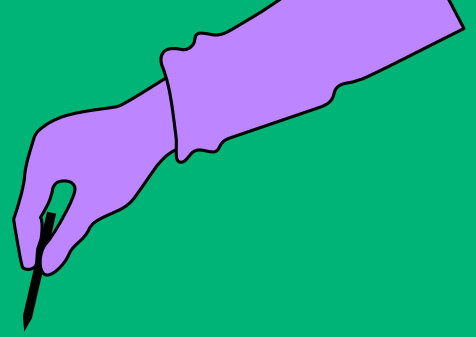
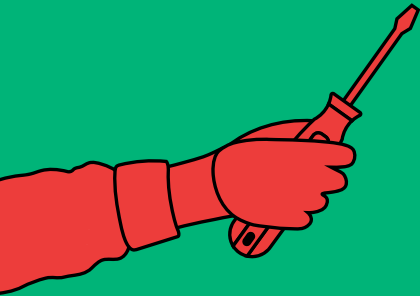


OPEN REPAIR ALLIANCE



THE RISE OF COMMUNITY REPAIR



THE PEOPLE AND THE DATA
CREATING A MOVEMENT



Credit: Mark Phillips / The Restart Project

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This report is produced by The Restart Project on behalf of the Open Repair Alliance.

It was written and compiled by Neil Mather and Ugo Vallauri, with support from Fiona Dear. It was designed by Anna Negrini.

We would like to give a huge thank you to Monique Szpak for her fantastic work in producing the Open Repair Alliance datasets over the years.

Special thanks also to Thomas Opsomer for all of his

support and advice since the beginning of the Open Repair Alliance.

Finally: to all the groups around the world running repair events, and to all the people trying to make their items last longer - thank you!

Published October 2024

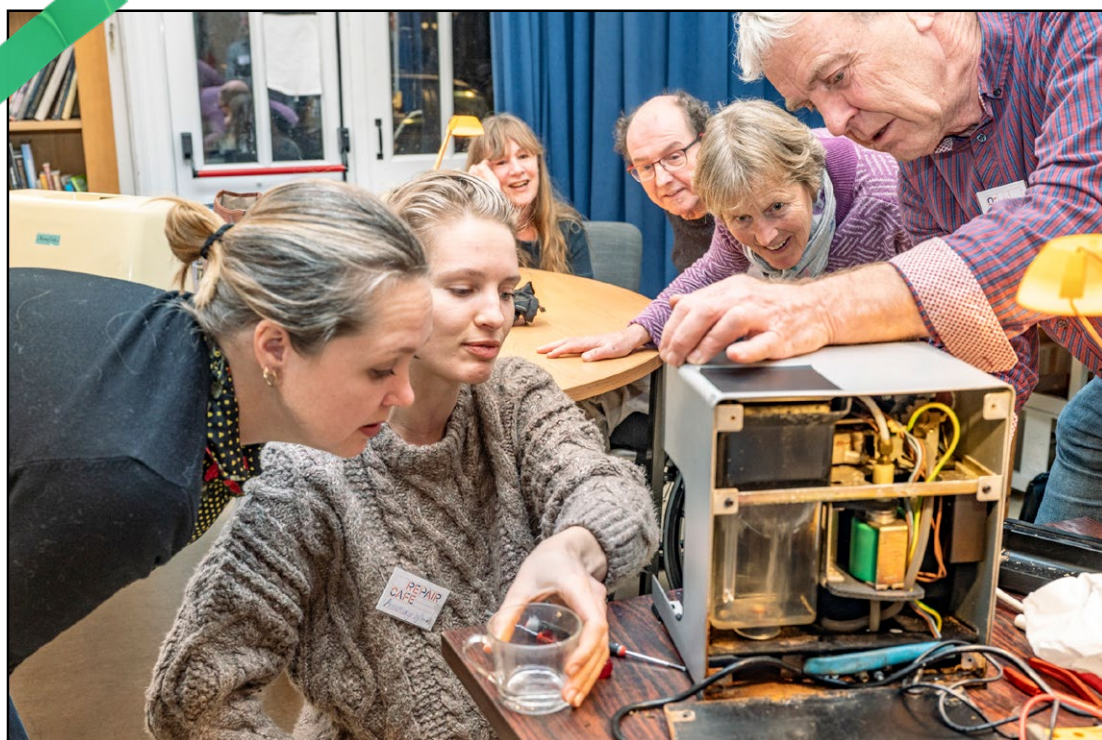
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EXECUTIVE SUMMARY

The Open Repair Alliance is an international collaboration of community repair networks. Our members help to organise thousands of repair groups around the world.

The concept of community repair is simple but powerful: gather together in your local community and work together to fix broken things. Since the start of Repair Cafés in 2009, every month, all around the world, community repair groups have been doing just that. And since 2017 the Open Repair Alliance has been pooling together the data that groups collect on the repair attempts of electrical and electronic products that they undertake.



Credit:
Martin Waalboer /
Repair Café International

In October 2024 we're celebrating 15 years since the first Repair Café. **Our dataset now includes 208,491 repair attempts documented by 1,158 groups in 31 countries during 19,986 events.** These numbers are impressive, but they vastly underestimate the true size of this movement. **We estimate that over 4,000 groups are active globally, performing 190,000 successful repairs per year.**

We and our partners use this citizen-powered data to make the case for the importance of repair and the right to repair. Insights from community repair data have been used in policy consultations and campaigning to help strengthen the case for a universal right to repair.

While there's been a lot of enthusiasm for progress in right to repair legislation in Europe, our data helps to put in perspective the huge remaining gap: **for 96% of all products brought to community repair events there's no right to repair in sight.**

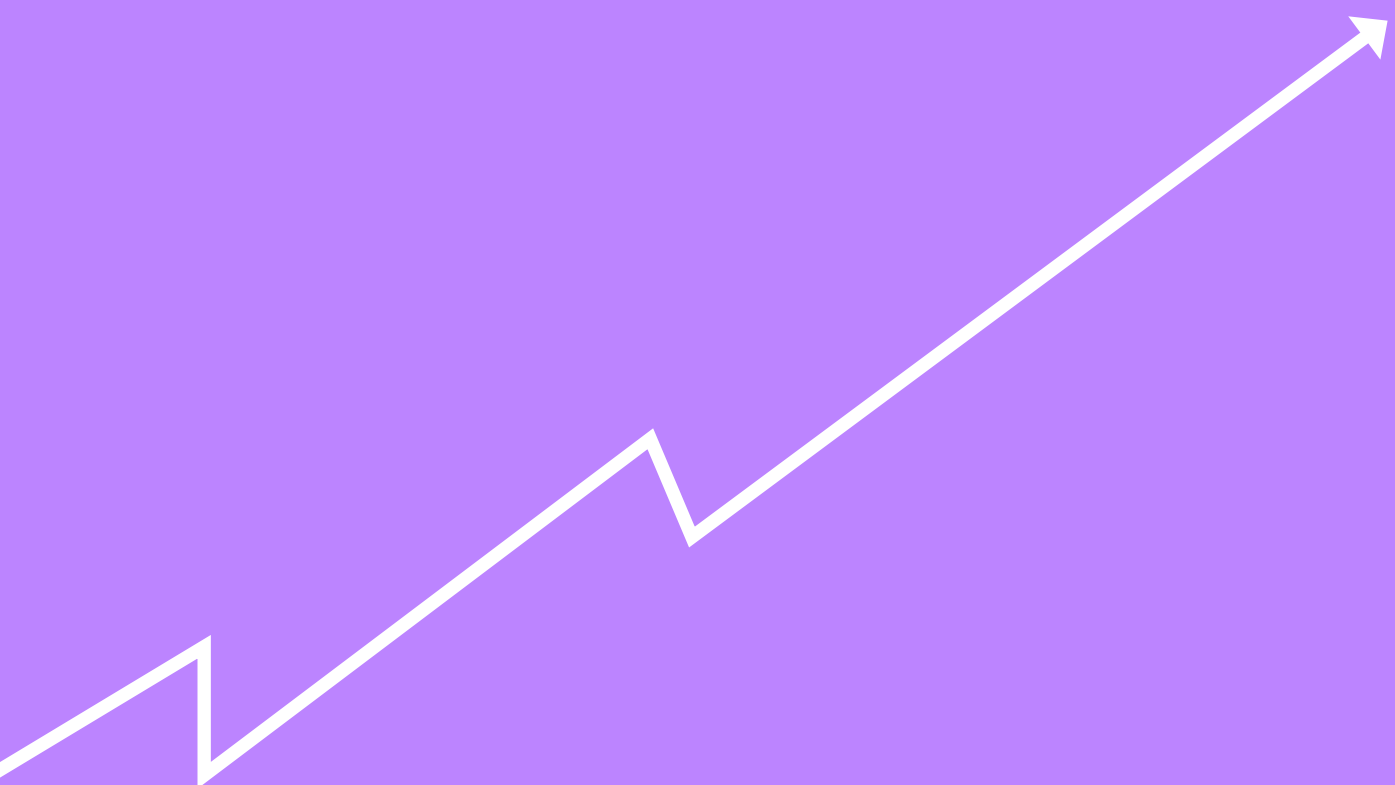
With the help of volunteers, participants in community repair initiatives are able to fix 53% of all products. This is impressive, as for many of these products there wouldn't be any commercial repair option. Yet, 25% of all attempts result in a product being deemed end of life, ready to be recycled. The main barrier preventing the completion of a repair is telling: **in 25% of cases it's lack of available spare parts, in 18% of cases parts are too expensive.** Product design plays an important role: in 16% of all cases, it's impossible to perform a repair because it's not possible to open a product. And in 12% of cases, repair information to complete a repair is missing.

Another important insight of the report is the average age of products seen at events. While mobile phones seen at events are on average less than 5 years old, products from all other categories are much older: 7 years for example for coffee makers, laptops and printers, 10 years on average for hair dryers and power tools; 18 years for Hi-Fi integrated systems, and a remarkable 28 years for projectors.

These insights highlight that **policymakers should consider much more ambitious measures to support repair and reuse of all existing products**, including financial incentives, and aim to **require manufacturers to support future products much longer than the 7-10 years included in current right to repair regulations.**

While regulation is slow to come, the role of community repair groups cannot be underestimated. As both the movement and the data it generates grow, we're also seeing the promising emergence of models bringing the experience of Repair Cafés to more permanent high street locations, we hope this leads to a return of a strong repair economy.

THE GROWTH AND IMPACT OF **COMMUNITY** **REPAIR**



THE GROWTH OF COMMUNITY REPAIR

All around the world, every month, people get together in their communities and do a simple but powerful act: they fix broken things. Since the first Repair Café in 2009, the idea of repairing things together in your community - 'community repair' - is now well-established.



Credit: Mark Phillips /
The Restart Project

Credit:
Martin Waalboer/
Repair Café International

15 YEARS OF REPAIR CAFES

The Repair Café concept was initiated by Martine Postma, who in October 2009 organised the very first Repair Café in Amsterdam. It was a great success and the concept has taken off around the world.

I was a journalist writing about sustainability and more specifically about waste reduction and I wanted to do more than 'only' write about this; I wanted to actively help change people's throw away behaviour. So I started thinking about why we create so much waste in our daily lives, and one of the things I found was that we no longer make repairs, people no longer know how to do that, they don't learn it anymore, not from their parents, not at school, it is not seen as an attractive activity, it's not held in high esteem. Buying new products on the other hand is considered attractive, fashion and advertising stimulate this behaviour, so we have gotten more and more focused on new products instead of repairing old ones. I wanted to change that, by showing people how much fun repairing can be and that it need not be difficult when you do it together with an expert volunteer.

How has it grown? We got a lot of media attention from the beginning, so the idea has spread via the media very fast. First in the Netherlands, then abroad. In 2012 we had a reporter over from the New York Times - and after that, from all kinds of media all over the world. Why was it so popular? I think because everyone deep down realises that repairing is the right thing to do, and that our throw away behaviour is not normal. So when an alternative to throwing away presents itself, people are happy to use that alternative. This gives me hope and stimulates me to go on spreading the concept and giving more and more people the opportunity to get broken items repaired.

Another reason for the growth of the movement is, I think, that Repair Café can be a solution to a variety of problems. There are different motivations for starting a Repair Café or visiting one or volunteering at one. People start Repair Cafés because they want to do something for the environment, or for the community, or they want an extra activity for their community centre, they want to attract new visitors, they want to do something for older people, or for younger people, or specifically for men, or for women. Or their religious faith is their motivation; they may want to protect God's creation. For repairers a reason may be to meet new people from the community, to share their skills, to stay active after retirement, to be a valued member of society, to learn from other repairers, to exchange skills, to have a good time, etc etc. So there are multiple reasons for starting or visiting or participating in a Repair Café - that's why it speaks to various groups of people and has more opportunity to spread.



Alongside Repair Cafés, there are similar concepts of community repair across the world - from Fixit Clinics in the US with a specific emphasis on collaborative repairs, to the electronics-focused Restart Parties which started in London, UK. Event organisers are also coming up with other names for these activities: from Caffé Riparazione in Switzerland, to Fiksefest in Norway, to Dr. Laptop in London.

I noticed that many common consumer items were getting harder to open because they used speciality or security fasteners. The first Fixit Clinic was an attempt to make uncommon disassembly tools available to people to see if we could open things up as the first step in troubleshooting them and fixing them. Fixit Clinic number one was held on December 1st 2009 at University of California Berkeley's Albany Village Community Center.

→ Peter Mui, Fixit Clinic



Credit:
Peter Mui / Fixit Clinic

THE OPEN REPAIR ALLIANCE

In 2017, at the very first FixFest in London, the formation of the Open Repair Alliance was announced by founding members The Restart Project, Repair Café International, anstiftung, and Fixit Clinic, with technical support from iFixit. Since then, we've worked hard to aggregate and share our collective data on repairs of electrical and electronic products attempted at community events around the world.

While we repair all kinds of items in our initiatives, ORA members chose to concentrate on these products due to concerns about the growing amount of electronic waste generated, and looking at future opportunities to influence product design and policies around repairability. These repair attempts tell an exciting story of the repair movement - every single record in our data speaks of people taking time to resist obsolescence and work together to fix their broken things.



Credit: Mark Phillips /
The Restart Project

We founded The Restart Project in 2012, inspired by our time working in the Global South and witnessing how people kept their things going – often out of necessity. Seeing this in contrast to the growth of electrical waste, overproduction and overconsumption in the Global North pushed us to bring that approach to the London community. We were surprised by the enthusiasm we saw at that first event in a pub! We figured out that it was entirely possible, doable, and actually quite exciting and fun. Our aims have not changed very much in the last decade - we have only scaled the level of ambition and found other avenues through which to bring change. While the community events and activities that Restart was founded on are still key to our strategy, so is Right to Repair campaigning supported by data insights we generate, and system change at scale.

→ Ugo Vallauri, The Restart Project

Almost 40 community repair projects existed in Germany in 2013, when anstiftung began supporting repair initiatives and connecting the projects. In autumn 2014, the first nationwide networking meeting took place at the Deutsches Museum Verkehrszentrum in Munich, followed by others in Berlin, Bielefeld and Kassel. People are now repairing things together, on a voluntary basis and in public at over 1,500 locations throughout Germany.

Almost every day, at least one repair event takes place somewhere and new initiatives are still being set up in many places. So that not every Repair Café has to reinvent the wheel every time, the Netzwerk Reparatur-Initiativen (Network of Repair Initiatives) is the point of contact for all questions relating to community repair initiatives.

→ Tom Hansing, Ina Hemmelmann, Linn Quante, anstiftung

WHY REPAIR?

For one, electrical waste (e-waste) is one of the fastest growing waste streams globally. Out of the 62 million tonnes of e-waste produced in 2022, 33% (20.4 million tonnes) were small devices, the most common products seen at community repair events¹. Only 12% of these are recycled globally. While recycling rates are higher in Europe, repair can help to cut this waste by keeping items working longer.

Why *community* repair? Along with repair businesses, community repair events are a crucial part of repair infrastructure. We fix things in Repair Cafés that often can't be taken for repair anywhere else, and would otherwise be thrown away. Community repair helps to foster a culture of repair, in response to the still dominant throwaway economy.

Community repair is growing, with more and more repair attempts logged over time. People don't want to throw away their broken devices. Volunteers around the world are helping to fix them. And if some key barriers to repair were removed, we could fix even more.

¹These stats are from the [Global E-Waste Monitor 2024](#) by Unitar.

Repairing is not a new everyday practice. But it is only since 15 years that people have been increasingly joining together in repair meetings, Repair Cafés and other repair initiatives in order to counteract rapid and planned obsolescence. Repair Cafés address the meaningless production of ever new things and devices. At these events, repairing is a conscious rejection of a culture in which products are already devalued by the next new generation of products as soon as they are put into use. By gaining an insight into the devices, volunteers and participants also want to reduce their dependence on experts and find out for themselves how things work. Their motto is: If you can't open it, you don't own it.

Repairing could therefore be the prelude to a truly more sustainable life. Since the logic of growth has long been inscribed not only in economic conditions, but also in the people - prosperity, progress and development are understood to mean a permanent increase - the challenge is to develop new models of prosperity that increase the quality of life, even though - or because - they limit material demands. This is where repairing comes in. It is an attractively wilful action that also strengthens the autonomy of individuals and communities.

While being a consumer promotes a carefree attitude towards the world (you bought the thing, you use it as long as it fulfils its purpose, you throw it away), when repairing together you ask about the origin of things, their life cycle and the consequences of their production and, despite the divisions mediated by the (world) market, you try to reconnect and take responsibility for the consequences of your actions - for extractivism and exploitative production conditions.

What happens when things are repaired collectively goes beyond the repaired toaster or the mended computer; the social experience of togetherness and social effectiveness are also crucial. Other ways of doing business can be directly experienced here: through co-operation, peer production, self-empowerment and commoning. Such experiences undoubtedly also serve to repair the (social) world. At the same time, collaborative repair creates social networks. Where there was isolation, there is suddenly a common interest, a collective action. As a temporary infrastructure in the neighbourhood, a repair initiative regularly brings people together in conversation, action and relationships. In the context of a local repair culture, repairing also creates cohesion. Repair events enable experiences of self-efficacy and thus also favour - or are themselves - a democratic innovation: here, the participatory shaping of social environments is attempted.

The question of the “good life” and a different approach to nature are on the agenda, and it is therefore logical that, whether in community gardens, open workshops or Repair Cafés, people rely on the subsistence logic of cooperation instead of the (capitalist) logic of competition. It’s also logical that they share knowledge (instead of monopolising it), and that they counter the industrial tendency to produce waste and obsolescence with the principle of giving things a longer life through repair and upcycling.

→ Andrea Baier, anstiftung²

² Shortened and adapted from [Reparieren Projekte, Orte und Akteure einer Bewegung](#).

Credit: Lauren-Victoria Mc Kown /
Café kaputt, leben.lernen.leipzig e.V.

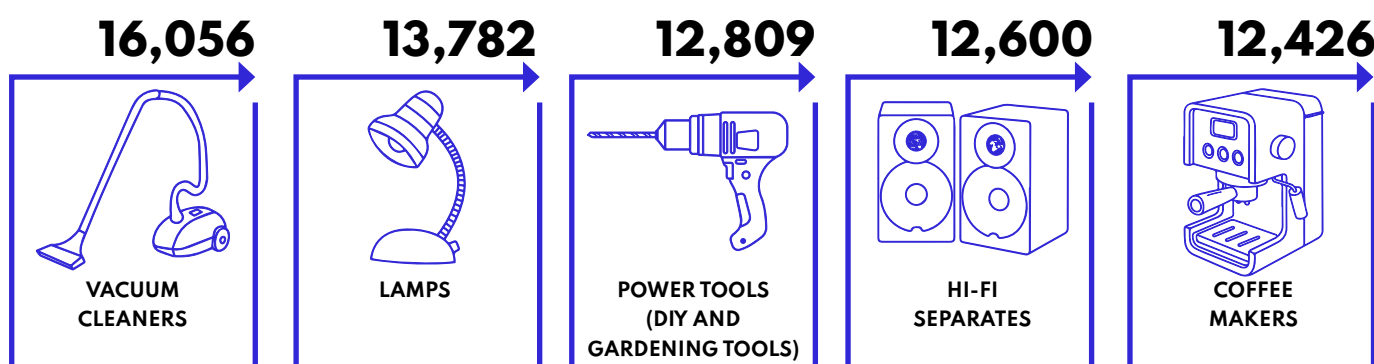


WHAT THE DATA TELLS US

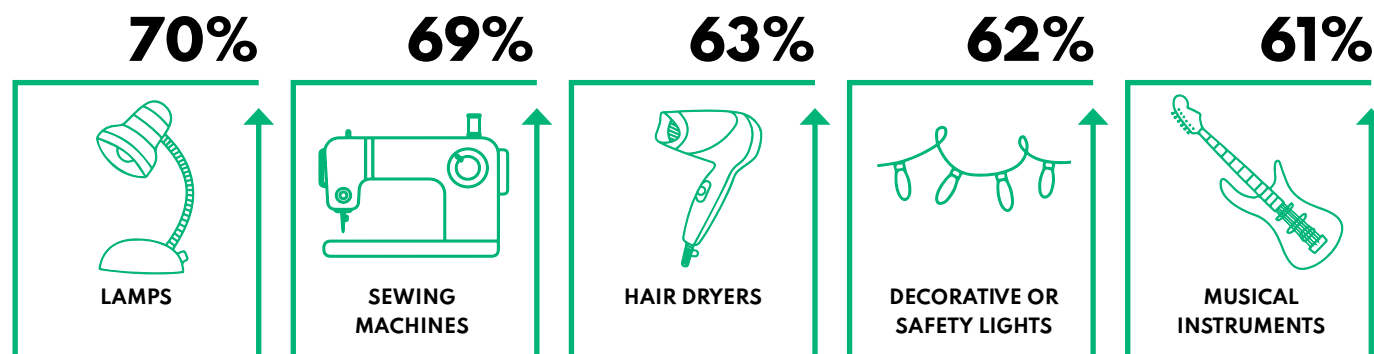
In the most recent set of Open Repair Alliance data, there are:

- 208,491 repair attempts of electrical and electronic items
- from 19,986 events
- run by 1,158 community repair groups
- across 31 countries around the world

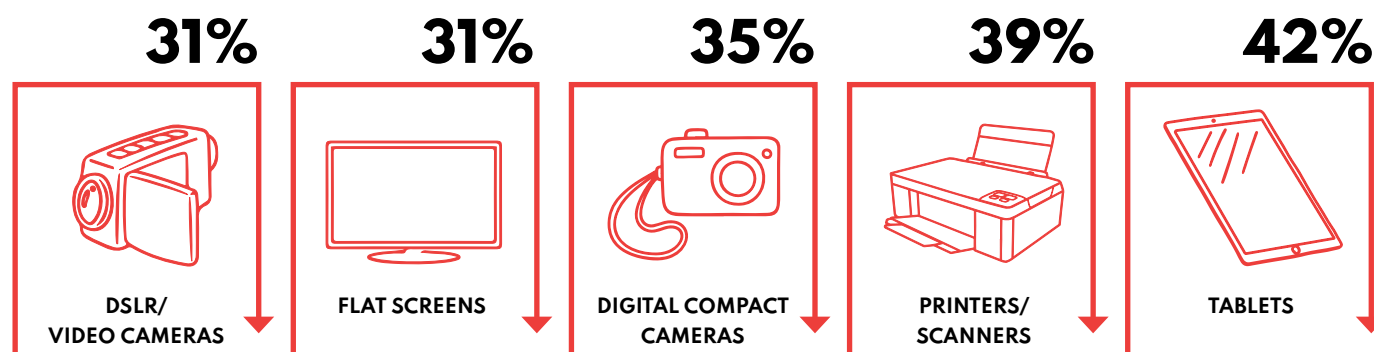
TOP 5 PRODUCT CATEGORIES SEEN MOST FREQUENTLY AT EVENTS



HIGHEST SUCCESS RATES



LOWEST SUCCESS RATES



³ Many Repair Cafés log repairs on paper but don't digitise these for online sharing.

⁴ Based on relative proportions of electrical and non-electrical products in Repair Monitor and Restarters.net.

⁵ Based on the number of repair groups registered with repaircafe.org, restarters.net, reparatur-initiativen.de and other data providers, accounting for overlaps and inactive groups. This is likely an underestimate of the number of repair groups globally.

⁶ Taking into account the average success rate of 53%.

Our dataset continues to grow at an increased pace. In the 12 months from August 2023 to July 2024, the Open Repair Alliance logged nearly 70,000 repair attempts of electronic and electrical items.

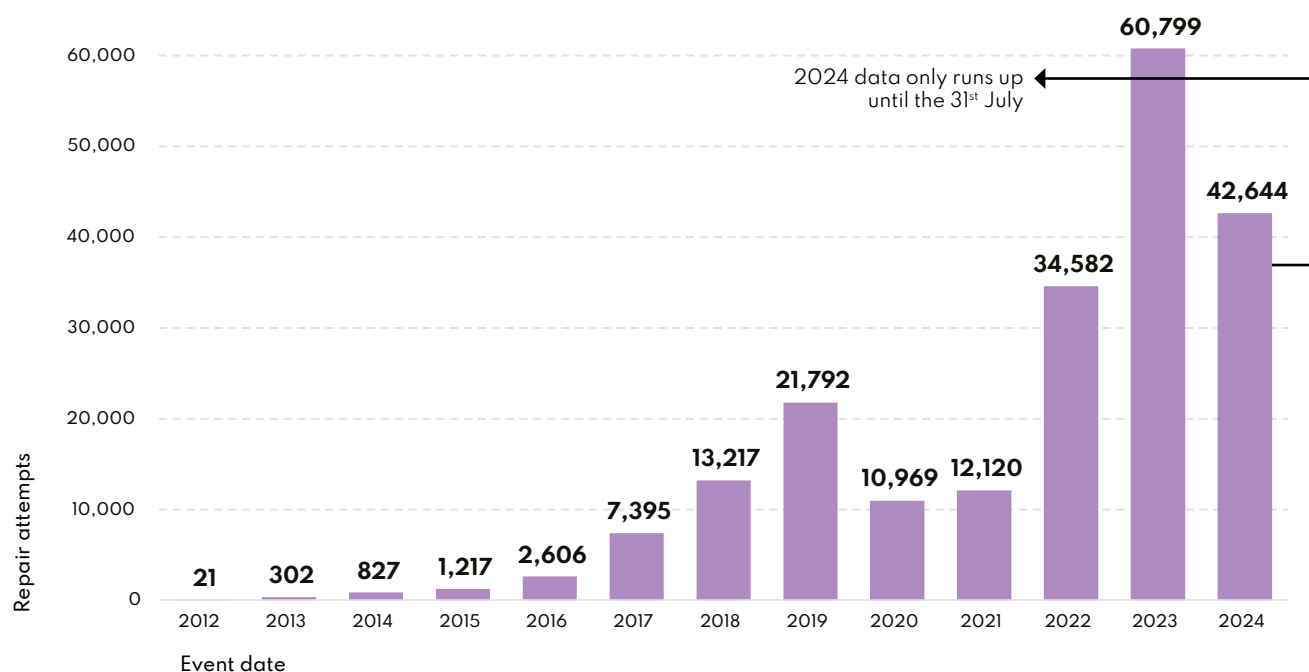
→ That's nearly 6,000 items logged a month

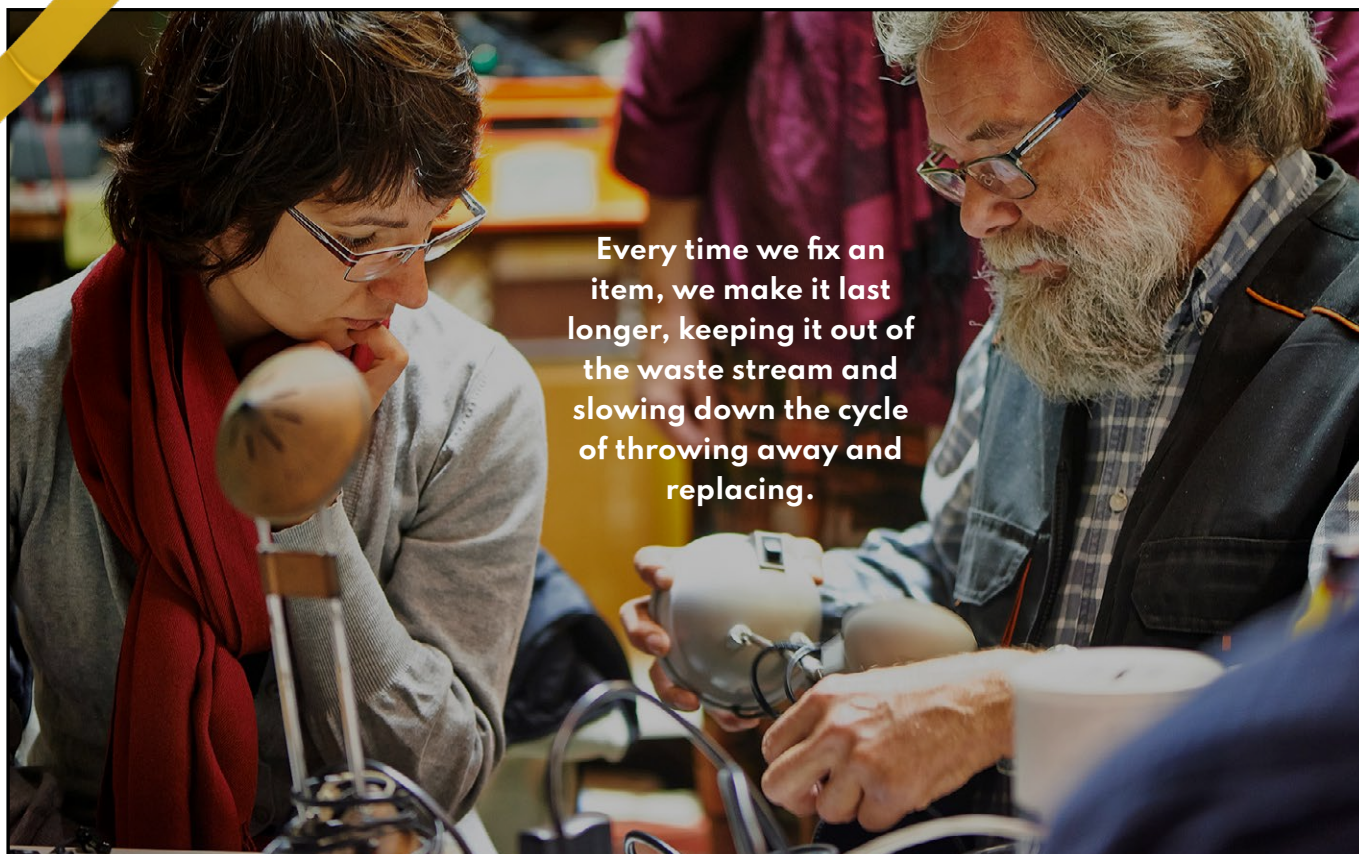
→ An average of nearly 200 repair attempts logged every day!

And all of this is only a small slice of community repair - it represents only the repair attempts of electrical/electronic items, and only those that have been logged and shared³. The true scale of repair community activity is much bigger! For instance, we know that for groups collecting data, non electrical products seen at events are 30-40% of the total⁴.

Using this data, we can estimate the scale of electrical repair attempts in community repair events across the world. Assuming there are at least 4,000 repair groups⁵ around the world, this dataset captures only around 20% of the total electrical repairs performed at community events. **This means there are likely to be approximately 30,000 repair attempts each month, or around 1,000 per day. And 190,000 successful repairs every year⁶.**

Repair attempts logged over time





Every time we fix an item, we make it last longer, keeping it out of the waste stream and slowing down the cycle of throwing away and replacing.

Credit: Lauren-Victoria Mc Kown / Café kaputt, leben.lernen.leipzig e.V.

Repair success rate



18%

“repairable” after the event



53%

fixed during repair event

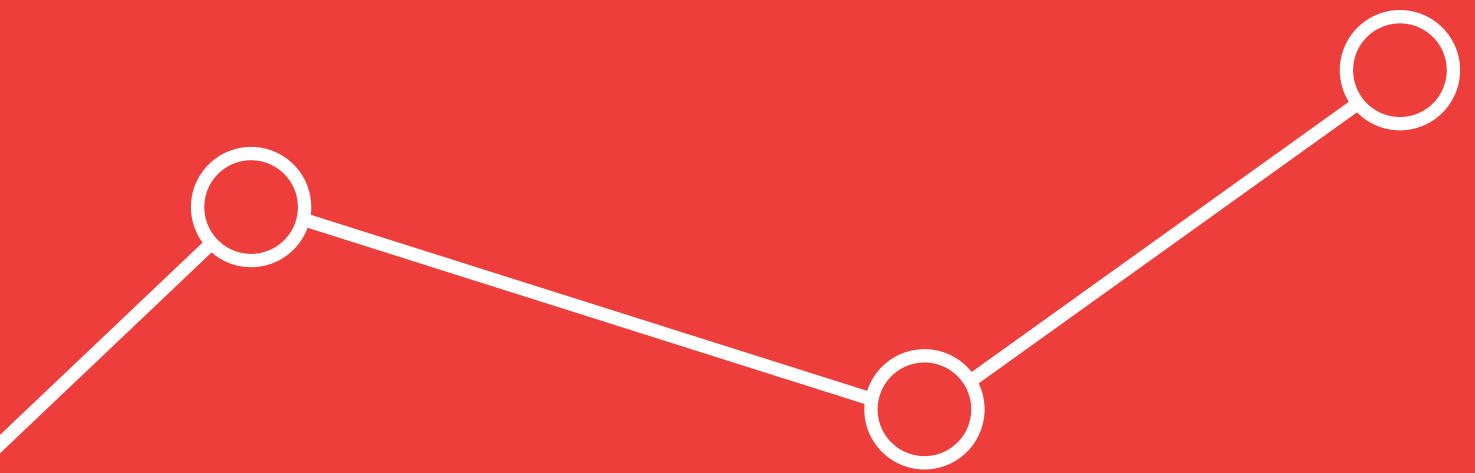


25%

end of life

On electronic and electric items we see a successful repair rate of just over 53% - already a big impact every month. But imagine if we could get that figure closer to 100%. What would it take? To date, 18% of products taken to repair events are deemed “repairable” but can’t be repaired because they need additional time, a spare part not available at the time, or additional skills. 25% of products are labelled end of life, ready to be recycled. By making all future products more easily repairable, extending access to repair information and affordable spare parts we could substantially increase the rate of products successfully repaired.

THE IMPORTANCE OF REPAIR DATA



WHAT IS COMMUNITY REPAIR DATA? AND WHAT DO WE USE IT FOR?

Every repair group is different. But over the years, many groups have realised that there's great value in recording data about the items they see, and their repair successes (or repair difficulties).

In 2017, the Open Repair Alliance identified core pieces of information being collected by repair groups that, when combined at scale, can make the case for the importance of repair - and the right to repair.

Key information recorded



Product category
What type of item is it?



Brand
Who made it?



**Product age /
year of manufacture**
How old is it?



**Fault/solution/
work carried out**
How is it broken?
How did we (try to) fix it?



Repair status
Could we fix it?



Barriers to repair
If we couldn't fix it, why not?

This information is supplemented with further data on event date, group, and country. See <https://standard.openrepair.org> for more information.

Collecting this repair data isn't easy - community repair events are busy, bustling activities where the main focus is on fixing something, not writing it down! As a result, for many of the records in the dataset we don't have complete information. Yet, our repair data is vital in telling the story of the importance of repair, and every piece of information we log helps to tell the story of the scale of repair happening in communities around the world.

It can also help to highlight the barriers that are stopping us from repairing even more. To help make the case for more repairable products, improved support and access to better repair services, we and others have analysed the data to help highlight the need and demand for repair. From right to repair policy consultations, to campaigning and activism, to journalism and academic research, the Open Repair Alliance data has been used widely.

IMPACT ON POLICY AND CAMPAIGNING

Over the years we have run multiple citizen science projects on the Open Repair Alliance data, using it to determine common fault types of key product categories. We've looked at mobiles, printers, tablets, batteries, laptops and vacuum cleaners.

Insights from these analyses have been used by the Right to Repair Europe campaign to make the case for spare parts and repair information to be made widely available.



Credit: Mark Phillips / The Restart Project

THE SHORTCOMINGS OF CURRENT RIGHT TO REPAIR LEGISLATION

Technically, almost none of the products we've seen at community repair events, in Europe or elsewhere, are covered by right to repair legislation. That's right: only 40 products out of the over 208 thousands logged so far, or 0.0002%. Why is that? Looking specifically at Europe, the only products already sold on the European market covered by ecodesign legislation with repairability requirements as of 2024 are models of white goods, welding equipment and TVs introduced on the market after 1st March 2021. White goods (such as washing machines,

⁷To date we've seen data logged for only 4 welders, without any information submitted on year of manufacture. As for white goods, counter-intuitively some people find ways to bring them to repair events: we've counted 150 in the dataset, although only 1 washing machine and 3 mini-fridges covered by EU legislation.

⁸ For more information on required improvements, see [Right to Repair Europe's analysis](#).

⁹ It's worth noting that we don't have any insight on whether these regulations will impact the repairability of products available in countries such as the UK and others not part of the EU single market.

¹⁰ Vacuum cleaners were mistakenly announced by the EU Commission as being covered by the upcoming Directive on consumers' right to repair, but repairability requirements for this product category haven't been finalised yet.

dishwashers and fridges) and welding equipment are very rarely seen at repair events⁷: only one washing machine and three mini fridges seen at events to date were covered by EU legislation. And while to date we've seen 1,866 TVs and monitors at repair events, only 36 of them were manufactured in 2021 or later.

Looking ahead, the new regulations on smartphones and tablets in the EU, which are projected to improve repairability and durability of future devices⁸, will only apply to brand new models introduced from June 2025. Even if we were to look at the Right to Repair Directive on consumer rights, which will be enforced in EU countries⁹ starting in August 2026, it will at launch only cover the same product categories: we recently learned that vacuum cleaners¹⁰ still don't have repairability criteria approved.

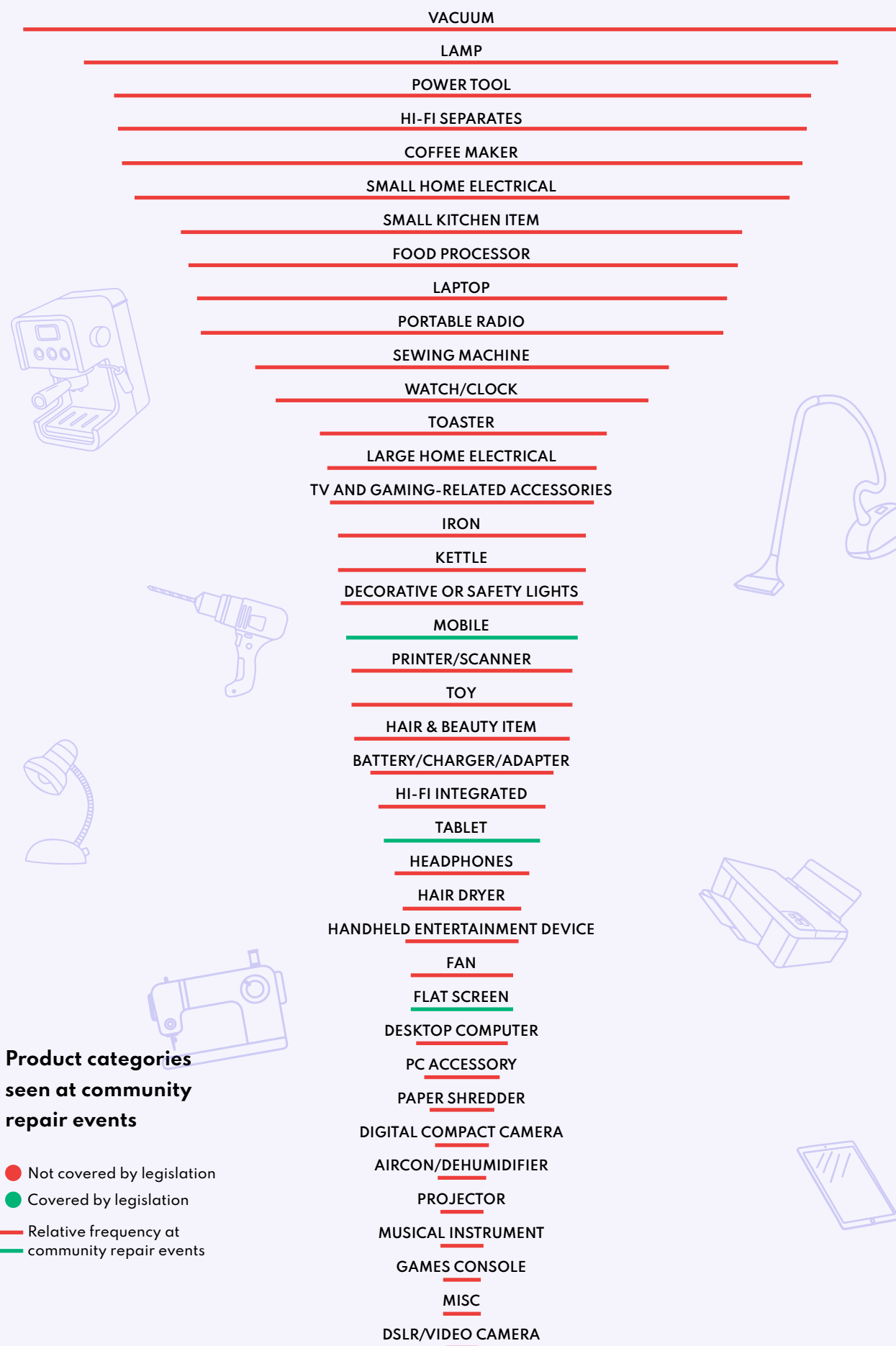
To date, we've logged just under 9,000 repair attempts of tablets, TVs, and phones. Out of the total amount, this represents only 4% of all items we've seen brought to community repair events.



Credit:
Brendan Foster /
The Restart Project

¹¹ Battery regulation in the EU will require manufacturers of most consumer products introduced from 2027 onwards to have user-replaceable batteries. But we need progress for all spare parts, not just batteries.

It is therefore safe to say that **for 96% of the electrical and electronic products seen at community repair events there's no right to repair in sight**¹¹. It is a shocking number that demands urgent action by policymakers.

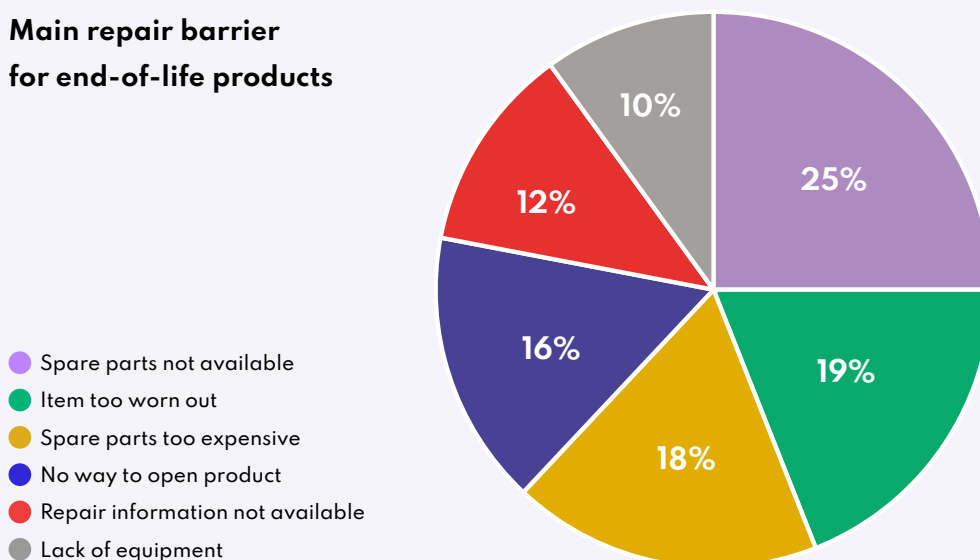


THE CASE FOR AMBITIOUS FUTURE LEGISLATION

BARRIERS TO REPAIR

When repairs are unsuccessful, and products are deemed end of life, we ask volunteer repairers to share the main barrier they've experienced: to date, we have 18,921 records (or 36% of all end of life products) with an indicated barrier. Our data shows that spare parts play an important role here - either by being not available during the repair session or on the market (4,676 instances, 25% of all barriers) or being perceived as too expensive, in 3,362 cases (18%). Often items are simply too worn out and no longer possible to repair. This was reported 3,617 times (19%): sometimes, simply looking after items better can help them to last longer.

**Main repair barrier
for end-of-life products**



Product design is also signalled as an important barrier: specifically when there's no clear way to disassemble a product to reach the part needing replacement, in 3,126 instances (16%). Repair and servicing manuals could also play an important role in reducing uncertainty for repairers in disassembling products they haven't previously worked on: Lack of available information was listed by volunteers 2,289 times (12%). Finally, lack of required equipment is a less prevalent barrier, experienced in 1,851 cases (10%).

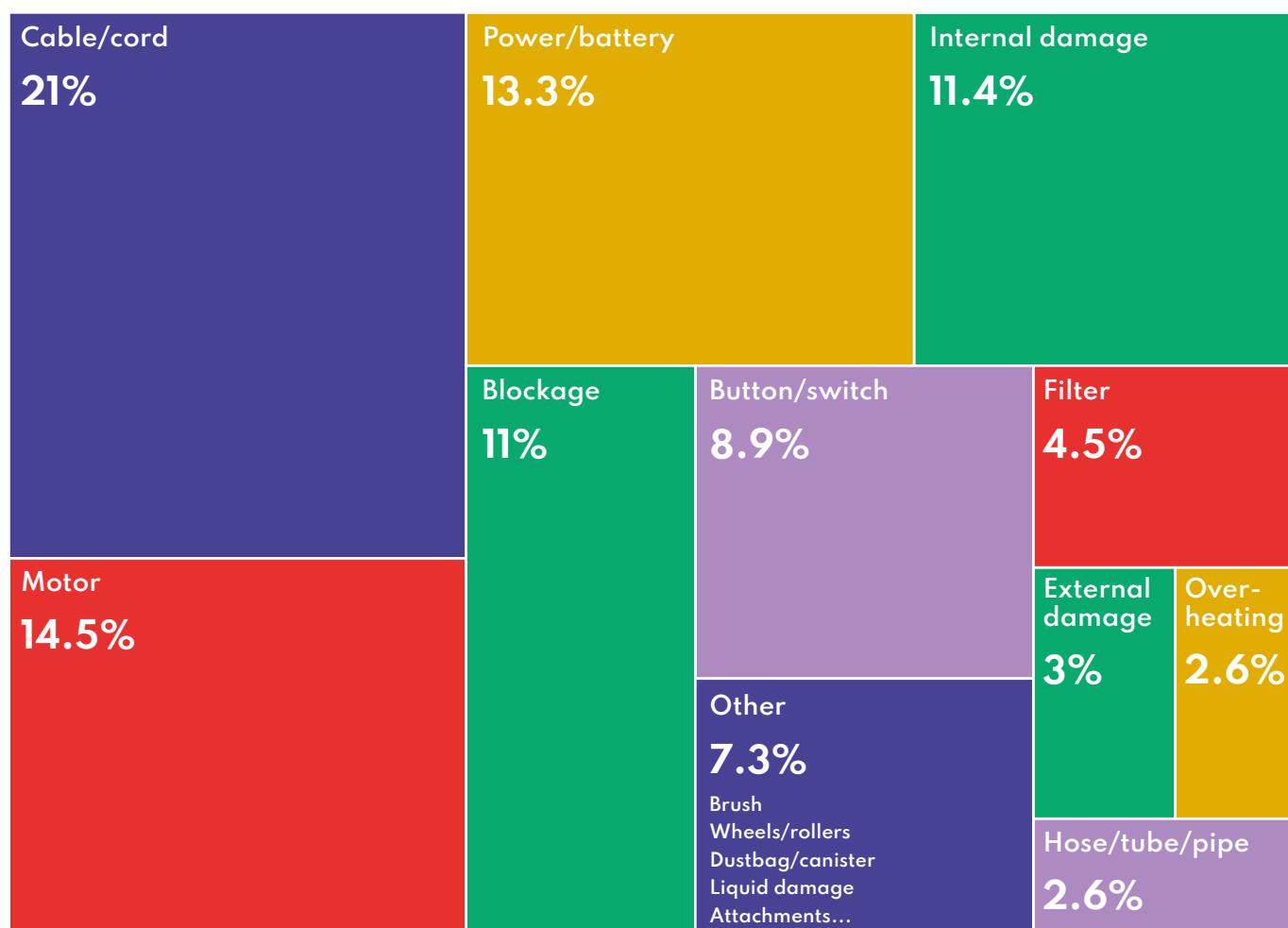
Ecodesign regulations that are slowly being introduced in the EU are addressing this aspect, but as we highlighted above, the vast majority of products are still not covered by repairability requirements.

AN EXAMPLE: VACUUM CLEANERS AT REPAIR EVENTS

Vacuum cleaners are one of the most popular items brought to community repair events - to date we've seen 16,056, or 7.7% of all products reported globally. The data we get from these repair attempts can be analysed to help generate useful insights on recurrent faults.

For example, in 2022, Open Repair Alliance partner The Restart Project facilitated a citizen science project¹² involving students from University of the Arts London and members of the repair community. Participants looked at Open Repair Alliance data on over 3,000 previous repair attempts of vacuum cleaners, specifically at the stated problem description, to identify recurrent fault "types".

¹² Full information on the methodology and raw data for this and other case studies can be found on the ORA's website, for example at: <https://openrepair.org/open-data/insights/vacuum-cleaners/>.

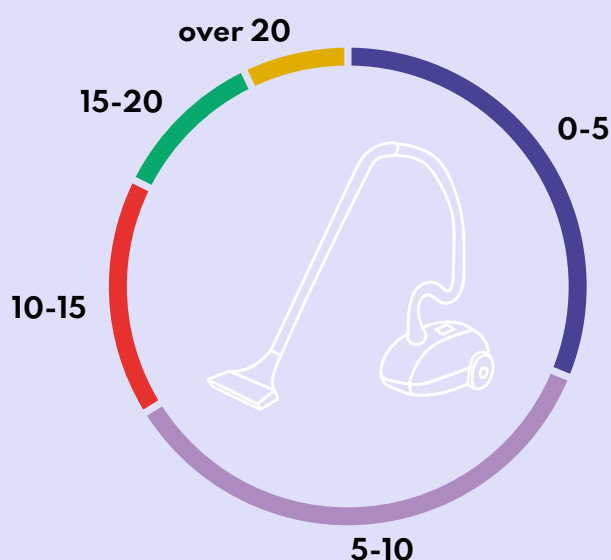


Most common vacuum cleaner faults

Data analysis demonstrates a wide range of faults experienced by volunteers and participants, as well as a very varying degree of success in performing a repair. For instance, more than 1 out of 5 faults seen at repair events are related to a cable or a cord of a vacuum cleaner, and they have one of the highest fix rates, at 81%

of all attempts. Motor faults are the second most frequent fault, at 14.5%, but by far the least likely to be repaired at an event (37%), due to the high price of spare parts. Blockage is another frequent fault (11%), with the highest success rate of repair at community events, 83%. This signals the importance of maintenance of products, such as good cleaning and understanding of how products operate.

Average age of vacuum cleaners in years



Credit: Martin Waalboer/
Repair Café International



This said, possibly the most important finding of the data analysis is participants' interest in keeping many products in use for far longer than manufacturers are currently prepared to support them for: of the products for which product age was submitted, we saw many aged in between 10-15 years (16%) and 15-20 years (11%), and even 7% beyond 20 years old, as well as 31% below 5 years and 35% in between 5-10 years¹³. Vacuum cleaners are still not covered by right to repair legislation in Europe - or elsewhere - and current draft regulations propose that some spare parts could be made available to professional repairers for only 5 years after retiring a model from the market, while some consumables and accessories might be made available to consumers for 10 years. Over a third of all vacuums seen at community repair events would not be covered by the legislation. The evidence from repairs performed by community repair initiatives is clear: **The proposed legislation is not enough if we want to extend the life of existing products in line with what many consumers want, rather than continue to promote recycling of serviceable products.**

¹³ These figures are based on records for vacuum cleaners repairs with product age in the dataset in 2022, at the time the analysis took place. 2024 figures are almost the same: 34% below 5 years, 34% between 5-10 years, 26% between 10-20 years, 6% beyond 20 years.

PRODUCT AGE AND IMPLICATION FOR RIGHT TO REPAIR

The example of vacuum cleaners is not unique: people taking part in community repair activities do all they can to keep products in use for as long as possible. We've collected data on estimated years of manufacture for over 69 thousand repairs, or 33% of the total, and the results are impressive and telling. While mobile phones seen at events are on average less than 5 years old, products from all other categories are older: 7 years for example for coffee makers, laptops and printers, 10 years on average for hair dryers and power tools; 18 years for Hi-Fi integrated systems, and a remarkable 28 years for projectors¹⁴.

¹⁴ Many of the projectors logged at repair events are slide projectors, thus explaining their longevity.

Average product age of products at repair events



¹⁵ For example, EU legislation on smartphones and tablets will demand that software and security updates are made available only for 5 years after a product is retired from the market. But what about “connected” features of devices like toasters, which are seen at community events on average when they’re almost 10 years old?

The data tells us two things that policymakers should take into account:

- Some products are experiencing faults well before being 10 years old. Currently many of these faults are only rarely tackled by commercial repair providers. Therefore legislation to increase post-sale support and repair services for a wide range of products is urgently needed.
- Products from a wide range of categories are kept and used for much longer than the (maximum) 10 years of support included in upcoming legislation. This means that we need to extend the availability of (affordable) spare parts beyond the traditionally accepted period of 10 years. Additionally, considering the rise in connected products, the implication of longer-term software support should be taken into account for future legislation¹⁵.

Additionally, the data highlights something obvious and yet little discussed: **Pushing for the right to repair should not be only about better future products. We should not forget the millions of products already in use.** While we can’t retroactively change the design of existing products, policymakers can help increase the chances of existing products getting repaired and reused, for example by establishing repair funds to remove barriers to repair. For many of the product categories featured in our dataset, it is increasingly difficult to find a commercial repair option, despite repair demand being much higher compared to what’s seen at community repair initiatives. Some initiatives are pioneering economic incentives for the repair of products, including both the costs for spare parts when the repair is performed at a Repair Café¹⁶, and a substantial reduction of the cost of a repair when choosing a professional repairer. We’d like to see this happen everywhere in order to slow down the cycle of premature disposal¹⁷ of products and generation of e-waste.

¹⁶ The recently introduced repair bonus in Berlin includes full reimbursement for spare parts costing over 25€ when a repair is performed at a Repair Café.

¹⁷ For a full review of existing repair funds, see [Right to Repair Europe](#).



Credit: Mark Phillips / The Restart Project

COMMUNITY REPAIR DATA IN THE WIDER REPAIR MOVEMENT

The Open Repair Alliance data is freely available for [download](#) for anyone to use. Since 2020, we've seen over 1200 downloads, with a wide variety of intended uses: from repair groups, to academics, environmental agencies, device manufacturers and news organisations¹⁸. Users downloading our dataset were trying to achieve multiple goals - from studying the market of repairs and spare parts, to investigating future product developments and researching the economic impact of the right to repair movement.

Key reports that have included Open Repair Alliance data¹⁹:

- Reparatur von Klima- und Einzelraumheizgeräten : Kurzstudie zur Relevanz des Reparierbarkeits-Index (R-Score). A decision-making aid to better assess the potential introduction of a reparability index for air conditioning and local space heating appliances within the framework of the Ecodesign Directive (Wuppertal Institut, 2024).
- Imaging equipment and its consumables. Preparatory Study for Ecodesign. Assesses the environmental impact of imaging equipment and identifies and evaluates design options with the potential to reduce the impact; also proposes ecodesign measures. (European Commission Joint Research Centre, 2024).
- Lifespans and the circular economy. Examines the potential benefits of lifespan extension across different product groups and explores the technological, economic, social and policy changes which can help realise these benefits in the UK. (UKRI Circular Economy Hub, 2023).
- Extending the life of consumer electronics. Provides estimates of the benefits of extending the lifespans of three widely used consumer electronics in the UK – smartphones, laptops and tablets. (UKRI Circular Economy Hub, 2023).
- An overview of Europe's repair sector. Provides an evidence-based update on the status of the repair sector in Europe. (European Environment Agency, 2022).
- Long Live the Machine: Discusses how ecodesign and energy labelling can prevent premature obsolescence of laptops (ECOS, 2020).

Credit: Lauren-Victoria Mc Kown / Café kaputt, leben.lernen.leipzig e.V.



¹⁸ And our data is also featured in a [course to help students learn data science](#).

¹⁹ For more, see our [Insights](#) page.

TESTIMONIALS

“The work of the Open Repair Alliance to improve the collection, sharing and aggregation of data across repair events has been invaluable in shining a light on community repair activities taking place across the UK and further afield. Testament to the value of the data being generated, I’ve not been able to write an article/output recently without citing it in some form!

OLIVER LYSAGHT,

UK Research and Innovation NICER CE-Hub

“The Open Repair Alliance dataset is a great resource to find out if repairs can succeed without undue attention to financial and time pressures. Studying the data set reveals real world problems. The insights gained should be incorporated into a repair-friendly product design.

CHRISTOPH TOCHTROP,

Wuppertal Institut

“The Open Repair Dataset is a powerful tool that demonstrates how social movements and activist spaces around repair are sources of knowledge and have indispensable expertise to understand and respond to current environmental challenges. The academic research that I develop in the field of STS, dedicated to exploring citizen and collective responses to environmental problems derived from technology, would not be thinkable or viable without this data and diagnostic reports.

BLANCA CALLÉN MOREU,

Autonomous University of Barcelona

“The Open Repair data platform has been an incredible source of information for our Repair Café. Whilst we collect some limited information locally, the ability to see how our small part contributes towards a much larger movement has been incredible for motivating our repairers. We display the information (combined with other sources) as a dashboard, displaying really interesting stats for repairers and visitors alike.

AMBER HOPPER,

Swansea City Centre Repair Cafe

“The Open Repair Alliance data offer valuable insights towards understanding reparability, product failure types and consumer willingness to repair. Such information can also inform market analyses that form a basis for product policy considerations.

CHRISTOFOROS SPILIOTOPOULOS,

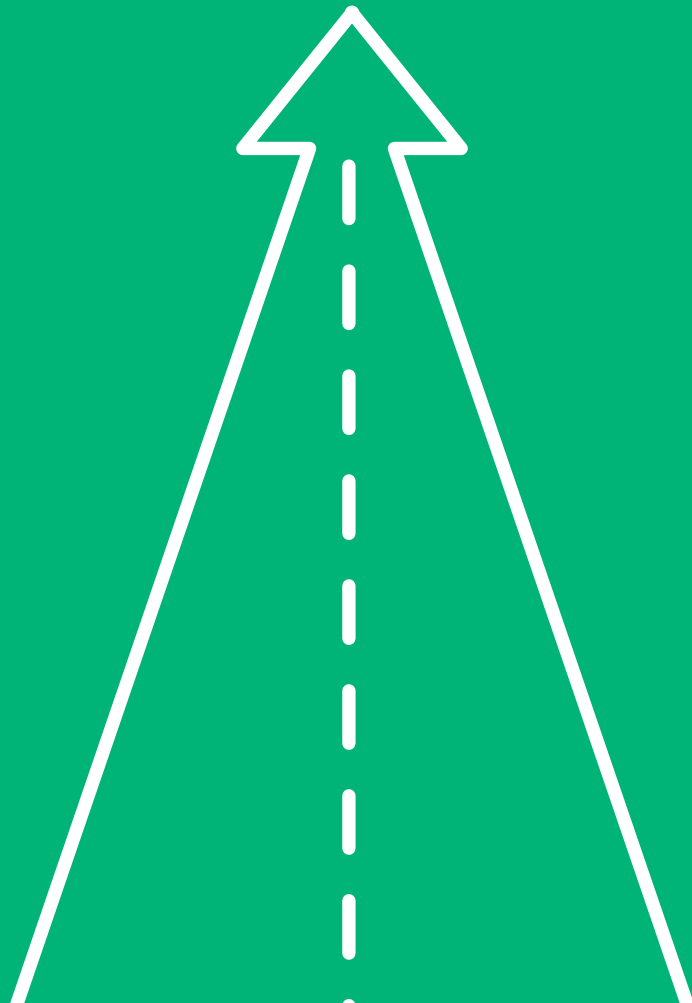
European Commission - Joint Research Centre

“The Open Repair Dataset is an indispensable tool for both researchers and educators. It provides rich, real-world data that is otherwise hard to access, enhancing our understanding of repair practices and serving as a key resource for informing the urgent need for policy reform in this area. Additionally, this is a valuable resource to embed in our teaching, and my students have found it incredibly useful to support their dissertation projects on various repair-related topics.

JAVIER LLOVERAS,

Postgrowth Innovation Lab - University of Vigo

FUTURE OF COMMUNITY REPAIR



WHERE NEXT FOR COMMUNITY REPAIR?

Over the last 15 years, the phenomenal rise of community repair groups has proven that repair is possible, and it's been an inspiration for many more people to choose to repair.

But the job of this movement is far from done! Things break every day, everywhere, and access to Repair Cafés - while massively increased over the last few years - is limited, leaving plenty of gaps to be filled if we want to fully move away from the throwaway economy.

In the UK as well as in other countries we're seeing the emergence of new models combining the collaborative nature of community repair events with a more constant presence to extend the visibility and reach of repair.

In 2022, The Restart Project and climate charity Possible launched two "Fixing Factory" pilots in London, creating spaces where people can bring in broken electronics and appliances to be fixed for free, while gaining the skills they need to make repairs themselves in the future. We trialled two types of Fixing Factory: one in a waste facility and one on a high street.

These were developed on the principle of community repair, but in a permanent space. The idea being that community-led repair could help to fill the gaps left by the disappearance of high street repair businesses. Whilst developing the concept, we had the pleasure of learning about the variety of different business models being developed by others that are opening community-led high street repair spaces elsewhere in the UK.

Fixing Factories - and other community-led high street repair spaces generate public interest in repair. They create a series of stepping stones to take people from never having looked inside an electrical item towards, if they wish, fixing for others. They are designed to empower local communities by teaching people how to repair their own stuff. They give broken products a second lease of life, saving people money whilst helping reduce carbon emissions by avoiding the need for new items.

Nearly two-thirds of people in London say there is a scarcity of nearby repair shops, and it's too expensive and hard to find other options. That's where community-led high street repair spaces come in.

Fixing Factories have already engaged nearly 1,000 people in community repair sessions, saving 1.5 tonnes of e-waste and 40 tonnes of CO2e over the last 2 years. And we can't wait for this type of initiative to bring back repair life to our high streets.

→ Fiona Dear, The Restart Project

REPAIR CAFÉS IN THE BROADER REPAIR ECOSYSTEM

Normalising repair - making it commonplace, ordinary and expected - is also about changing perspective. In the words of Fixit Clinic's Peter Mui, **it's about becoming an owner or guardian of products rather than a consumer** to emphasise stewardship and custodianship.

The curiosity generated by active participation in a community-led repair activity can help transform people's perception of what's possible and understand that repair should always be our first option when something stops working.

We also see an important opportunity for more collaboration and inspiration between Repair Cafés and independent repair businesses, which every day offer their services to save products otherwise destined for end of life. Community repair initiatives remove barriers by creating fun and engaging opportunities to rethink our understanding of what's possible, and to give everyone new skills and perspectives. This should extend to the whole repair economy, with appropriate support to make the repair ecosystem accessible and rewarding for all.

Community repair motivates people for repair and for a more sustainable lifestyle, because it makes them see with their own eyes that repair is possible. It shows them that repair is an option for their product too. By seeing that a repair need not be difficult and expensive but can be easy and affordable, they begin to understand that it's crazy to throw an entire product away when it's only a wire that needs to be reattached, or a filter that needs to be replaced.

This will motivate them not only to visit Repair Cafés, but will also focus their attention more on professional repairers - because they have become aware of the benefits of repair.

Repair Cafés and other community repair initiatives are not the sole solution to the problem of our throw-away society, but their success and the growth of this movement is a signal that more and more people begin to understand that our throw-away behaviour is a problem and that we need to change it in order to keep the world liveable for future generations.

In the sustainable future that we strive for, repair should be available every day everywhere in many different forms. We need people maintaining their products and making repair at home, we need Repair Cafés and professional repairers.

THE FUTURE OF REPAIR DATA

Real-world repairability data was hard to find before we began the journey of the Open Repair Alliance. We knew other data sources exist, but they're either inaccessible without paying large subscription fees, or very partial - for example telling the story of failure rate for just one type of product, often used in corporate environments. That's what's particularly refreshing about the work we've been aggregating and publishing over the years: **the Open Repair Alliance datasets makes available a lot of data that previously wasn't accessible at all.**

In discussions with manufacturers, we've also learned that often even they don't know what exactly happens to products beyond the end of their legal guarantee. If they sell spare parts, they can guess repair trends based on the popularity of the spares they sell, but they lack a lot of information. Similarly, repair businesses collect a lot of data, but don't share it publicly, and they tend to be more interested in the repairs they can perform, not in those they've decided not to offer.



Credit: Mark Phillips / The Restart Project

As the number of community repair groups continue to grow globally, this dataset will continue to expand. The standard behind this work doesn't limit data collection to community repair events, and we hope to start seeing more openness and sharing of data from other actors in the repair ecosystem: from manufacturers to repair businesses, spare parts distributors and insurance companies covering repair.

We've also seen the increased interest in downloading the data for further analysis. We hope to be able to collaborate further with partners to increase the value of the insights we can generate, and to make data analysis easier and more accessible for everyone.

CONCLUSION AND RECOMMENDATIONS

This report demonstrates the important role played by community repair in keeping hundreds of thousands of products in use that would otherwise end up as e-waste prematurely. The data we collect is a powerful tool to help visualise the change that needs to happen if we want to start reverting trends on e-waste and begin reducing it effectively.

We therefore ask policymakers at all levels to consider the implications of our work, especially the need to swiftly remove existing barriers to expanding repairability of products.

Regulations on right to repair need to apply to a much wider range of product categories, as our data highlights.

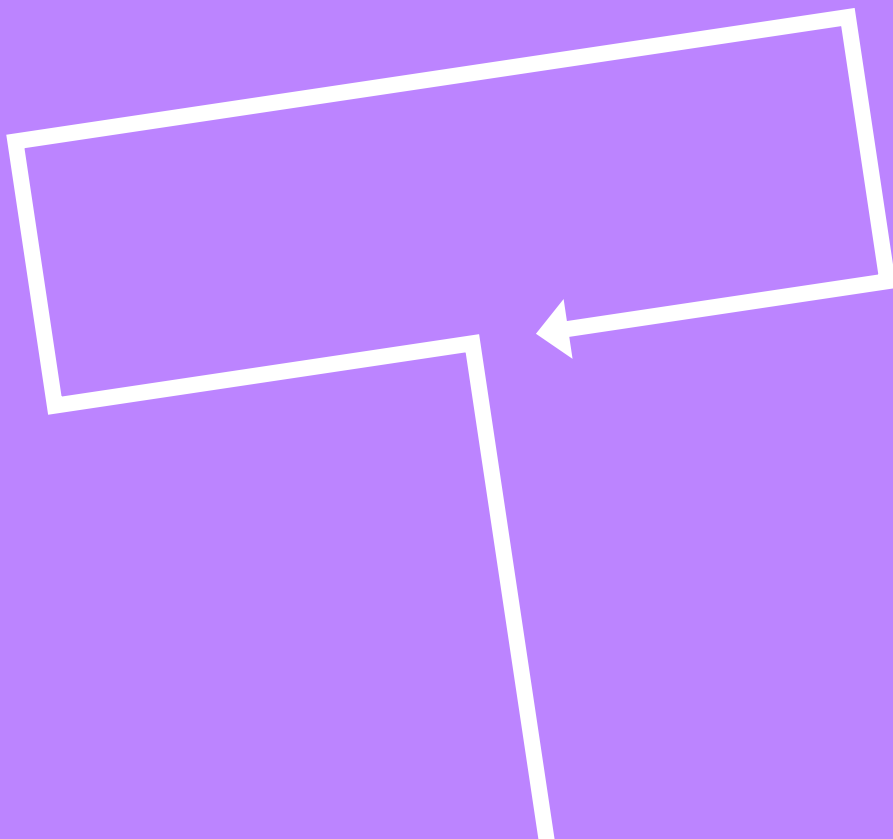
And **legislators need to think not just about future products, but also about ensuring that existing products are repaired instead of replaced**²⁰. Meanwhile, policymakers should also practically support local community efforts, helping to cover the costs of organising activities and training to increase their impact, while contributing to promoting and popularising the importance of repair at all levels.

²⁰ In addition to repair funds already available in Austria, France and part of Germany, the new right to repair law in Oregon, US, will require manufacturers to support devices produced as far back as 2015 (mobile phones from 2021) - see iFixit for more information.

We thank community repair groups for their contribution to the impressive achievements of this movement to date. For groups that contribute data, we recommend they **also use the aggregate data to help inspire further volunteers and participants at Repair Cafés globally**. In parallel we invite other groups not yet sharing their data to join their effort, so that we have a more comprehensive picture of barriers experienced, and more opportunities to influence policymakers at a local level too.

Finally, while the dataset shared by the Open Repair Alliance only features data collected at community events, our findings are important for the whole repair sector. We know that emerging social enterprises and startup repair businesses have downloaded the dataset to explore opportunities in the provision of new repair services. While we'll continue to collaborate with Right to Repair Europe and others pushing for better legislation to promote repairability, **we also invite repair businesses to continue to innovate to offer repair services** to help match the massive demand of repair solutions emerging from the community repair sector.

APPENDICES



OPEN REPAIR ALLIANCE MEMBERS



REPAIR CAFE



CORE MEMBERS

The core members provide repair data and help steward the direction of the Open Repair Alliance.

The Germany-based **anstiftung** Foundation carries out research into DIY culture and regional sustainability efforts, challenging the common definition of 'prosperity' as purely economic and drawing attention to the potential for local projects to improve quality of life on both regional and global levels. As part of their work, they coordinate the **Netzwerk Reparatur-Initiativen** national network of repair initiatives, involving Repair Cafés across Germany.

US-based **Fixit Clinic** holds pop-up community events where anyone can bring broken items to disassemble. As well as teaching valuable repair skills and helping to prevent unnecessary waste, Fixit Clinic has a strong emphasis on education: they strive to demystify science and technology and teach participants to think more critically about the way we consume and use our products. Their events are founded on the belief that a society where citizens are better informed about the technologies they depend upon is a society that is better equipped to collectively make important policy decisions. Started in California, its work has been replicated in Colorado, Minnesota and Massachusetts.

Repair Café International began with a community event held by Martine Postma in Amsterdam in 2009. It aims to maintain and spread repair expertise, and to promote social cohesion by bringing together neighbours from all walks of life and sets of motivations in the form of inspiring and accessible meetings. It now provides professional support to thousands of community repair groups worldwide.

The Restart Project is a people-powered platform for change, helping others realise the value of electronics and appliances, and use them for longer. Founded in London in 2012, The Restart Project works not only with communities and schools, but with academics, companies, government and like-minded organisations to highlight barriers to repair and longevity, and help drive a global movement to move away from the throwaway economy. Information about faults and repairs are collected at Restart Parties: free community events for learning electronic repair skills, held in London and beyond.

Core members worked with technical advice and support from iFixit Europe to establish the Open Repair Data Standard.

DATA PARTNERS

In addition to the founding members, a growing number of repair networks share their repair data for inclusion in the Open Repair Alliance data.



Cambridgeshire Repair Cafe Network coordinates Repair Cafés in the UK county of Cambridgeshire.



Hampshire Repair Café Network coordinates Repair Cafés in the UK county of Hampshire.



Repair Cafe Aotearoa New Zealand coordinates Repair Cafés across Aotearoa New Zealand.



Repair Cafe Denmark is a non-profit organization with repair initiatives all across Denmark.



Repair Café Hauts de France coordinates repair initiatives across northern France.



Repair Cafe Wales opens & supports Repair Cafés across Wales.



Repair Together represents the network of Repair Cafés in Brussels and Wallonia.

ADDITIONAL RESOURCES

- Data downloads. Download the Open Repair Alliance dataset.
- Supporting analysis and data for the facts and figures included in this report.
- Repair data collection tips. An overview of the different ways that repair groups can collect repair data.
- Producing the Open Repair Alliance datasets. Find out about the technical processes involved in combining the data from partners to produce the Open Repair Alliance datasets.
- Open Repair Data Standard. The full technical documentation of the data that we collect and publish.
- Repair data tools. A collection of tools, queries, data and code snippets for use with the ORA dataset.

Credit: Martin Waalboer/Repair Café International

